

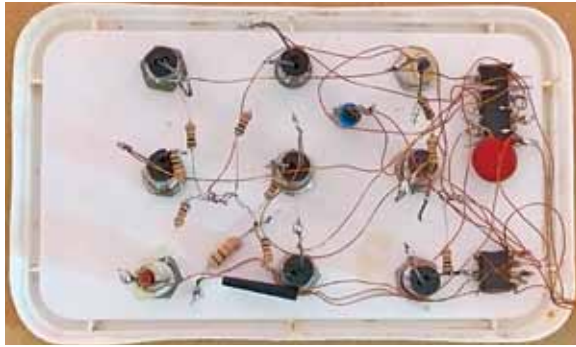


Fig. 3: Back side of the author's prototype

arrangement is very close to 50%. The series of square wave output pulses produced have a time period (T) equal to approximately $2(0.693) \times R \times C$. The output waveform frequency (f) is, therefore, equal to $0.722/R \times C$, where R is in ohms and C is in microfarads. The output of IC1 is further fed to the clock pin of binary counter IC2.

The 4020 IC is a 14-stage binary

counter, and we have utilised 9 outputs. Each output drives an LED in series with a resistor, producing different patterns before the sequence repeats. In the 4020 IC, the reset pin is connected to the ground with a $0.1\mu\text{F}$ capacitor. If not connected, the IC will reset repeatedly, resulting in

no output on the output pins.

Construction and testing

Before assembling the circuit, it would be beneficial to refer to Fig. 2 and Fig. 3 for the schematic circuit diagram and photographs, providing an exact idea about the construction of the device. Assemble the circuit directly in a suitable plastic box after connecting the LED holder to

the box. Mount the LEDs on the lid of the box using LED holders.

In a non-conventional approach, we may directly solder thin enamelled copper wires to various pins of the ICs without using a PCB or a veroboard. Subsequently, we may glue the ICs on the back side of the lid of the small plastic box and solder the other ends of the enamelled copper wires to the other components following the schematic. Each LED is connected through a current-limiting resistor (R2 through R10).

EFY note. Excessive heat from the soldering gun can damage the ICs, so exercise caution and work swiftly! If unsure, consider using IC holders. **EFY**



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